Spectral Unmixing for Crop Disease Detection

Concepts and methodology from five studies, with emphasis on how each one unmixes mixed pixels into healthy, diseased, soil and shadow fractions

Studies covered. Fitzgerald et al. 2004 (cotton spider mite, AVIRIS); Kumar et al. 2012 (citrus greening, airborne hyperspectral and multispectral); Zhang et al. 2014 (wheat powdery mildew, HJ-1 CCD satellite); Clenet et al. 2019 (vine disease, UAV, airborne and Sentinel-2); Guan et al. 2025 (peanut leaf spot, UAV hyperspectral). The document covers concepts and methods; it does not report results.

1. Concept: why unmix?

A pixel is rarely pure. In a 10–30 m satellite pixel, and even in a 5 cm UAV pixel at a leaf edge, healthy leaves, diseased leaves, soil and shadow are mixed. Mixed crop–soil spectra can even resemble disease spectra (Guan 2025), and mixture effects are strongest at satellite resolution (Clenet 2019). Spectral unmixing estimates the pure spectra present (**endmembers**) and the share of the pixel each covers (**abundances** or fractions); the diseased-crop fraction becomes a map of where, and how severely, the crop is affected (Fitzgerald 2004).

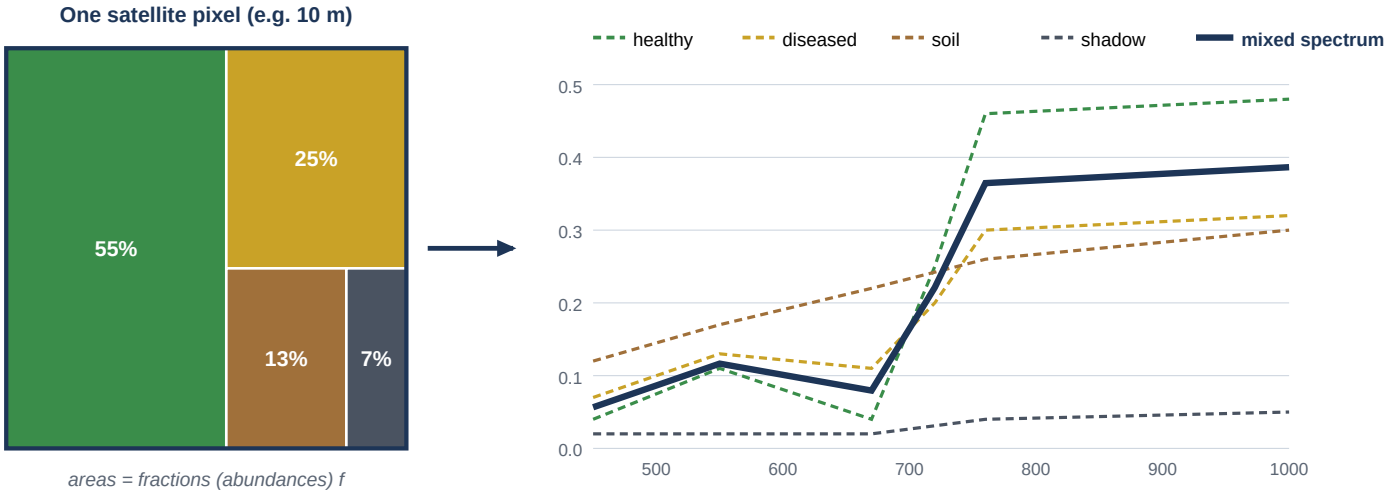
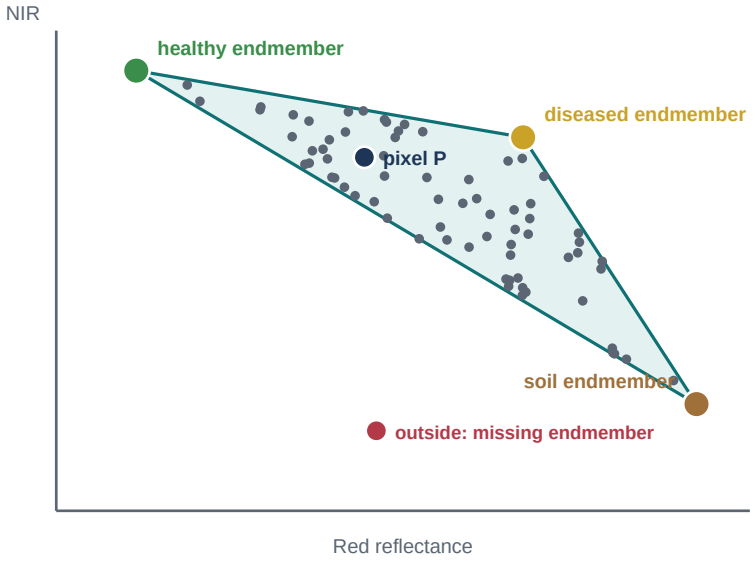


Figure 1. A mixed pixel and its spectrum: the observed spectrum (solid) is the area-weighted sum of the endmember spectra (dashed). Spectra are illustrative.



How to read it

Each endmember is a corner of the triangle (a simplex). Every mixed pixel lies inside it.

Pixel P sits nearest the healthy corner: **f_healthy = 0.5, f_diseased = 0.3, f_soil = 0.2.**

A pixel outside the triangle gives negative fractions: an endmember is missing or wrong *(Fitzgerald et al. 2004).*

With n bands, at most n + 1 endmembers can be separated, so 4-band sensors allow few.

Figure 2. Geometric view: endmembers are the corners of a simplex in band space, and fractions are a pixel's position inside it.

2. Key terms and equations

Term	Meaning	Used in
Endmember	Spectrum of one pure material (healthy leaf, diseased leaf, soil, shadow)	All five
Abundance / fraction	Share of the pixel area covered by an endmember	All five
Sum-to-one (ASC)	Fractions in a pixel add up to 1	Fitzgerald 2004; Guan 2025
Non-negativity (ANC)	No fraction below 0	Guan 2025 (FCLS)
Linear spectral unmixing (LSU)	Pixel spectrum = weighted sum of endmembers; solve for the weights	Fitzgerald 2004; Kumar 2012
MESMA	Each pixel may use a different subset (2–4) of endmembers from a library; keep the lowest-error model	Fitzgerald 2004
MTMF	Find one target (disease) against an unknown background; outputs MF score and infeasibility	Kumar 2012; Zhang 2014
FCLS / D-FCLS	Least squares with both constraints; D-FCLS uses only the endmembers nearest each pixel	Guan 2025
Non-linear unmixing	Handles multiple scattering between leaves and soil; here via overcomplete dictionaries and sparse projection	Clenet 2019
MNF, PPI	Minimum noise fraction (noise-sorted transform); pixel purity index (finds extreme, pure pixels)	Kumar 2012; Zhang 2014

Quantity	Equation	Source
Linear mixing model	$R'_i(\lambda) = \Sigma_{k=1..N}\, f_{ki} \cdot R_k(\lambda) + e_i(\lambda),\ \Sigma\, f_{ki} = 1$	Fitzgerald 2004
Model error	$RMSE_i = [\ \Sigma_\lambda\, e_i(\lambda)^2 \div n\ ]^{1/2}$	Fitzgerald 2004
Shade normalisation	$M_s = M \div (1 - Sh)$	Fitzgerald 2004
Fully constrained LS	$\hat{s}_j = \text{argmin}\ \lVert\, y_j - A_j s_j\, \rVert^2$ subject to $s_j \geq 0$, $\Sigma\, s_j = 1$ (A_j = nearest endmembers in D-FCLS)	Guan 2025
Homogeneity index	max over neighbours y of $[\ \lVert x - y \rVert + \cos^{-1}(x \cdot y \div \lVert x \rVert \lVert y \rVert)\ ]$ (Euclidean + spectral angle)	Guan 2025
Disease index	$DI = \Sigma(x \cdot f) \div (n \cdot \Sigma f) \times 100\%$	Zhang 2014
Two-date spectral change	$(F_{later} - F_{earlier}) \div (F_{later} + F_{earlier})$	Zhang 2014

3. Integrated unmixing workflow for disease

The steps the five studies share, merged into one sequence. Each box names the studies the step comes from.

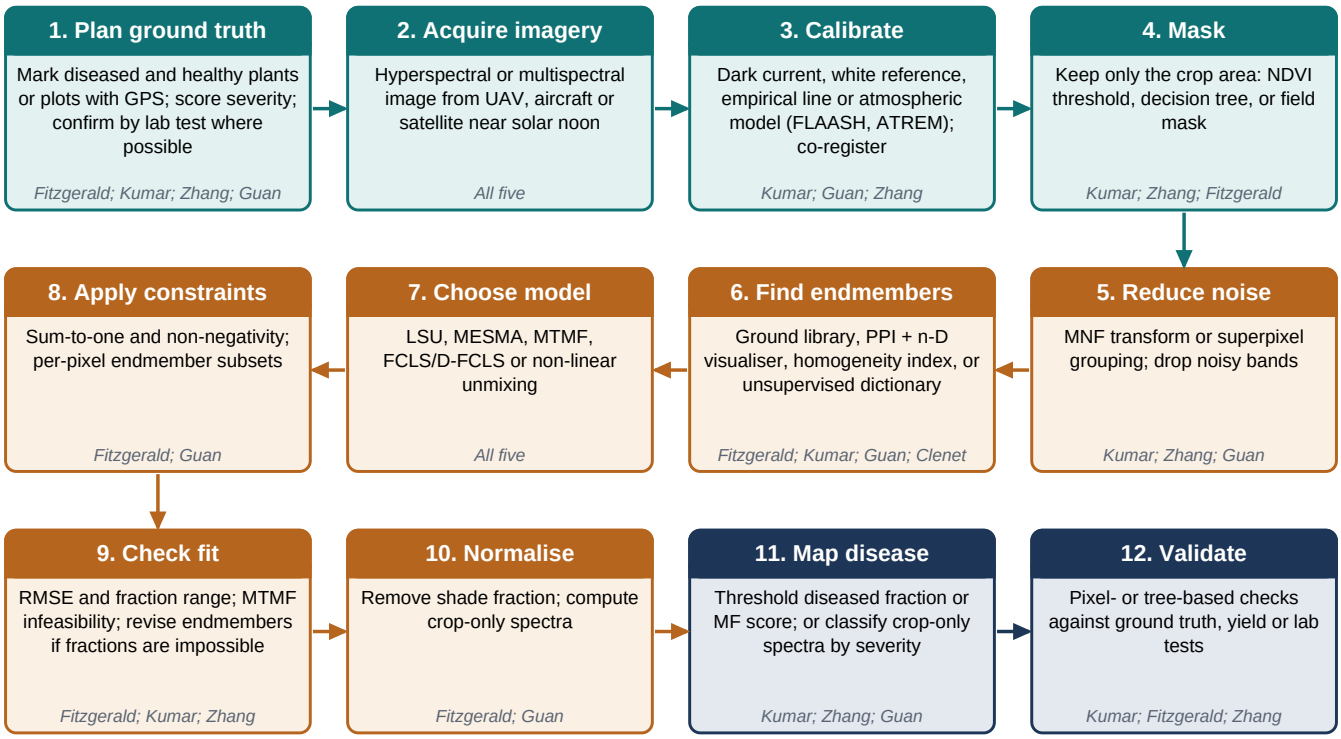


Figure 3. Integrated workflow. Teal: data preparation; amber: unmixing steps; navy: disease mapping and validation.

4. Finding endmembers: four routes

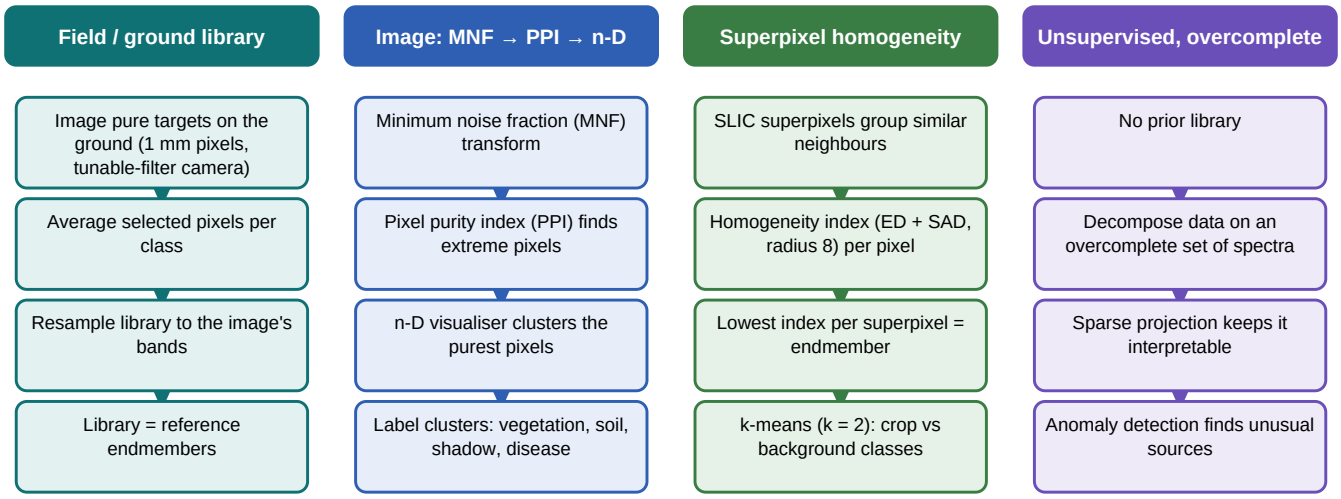
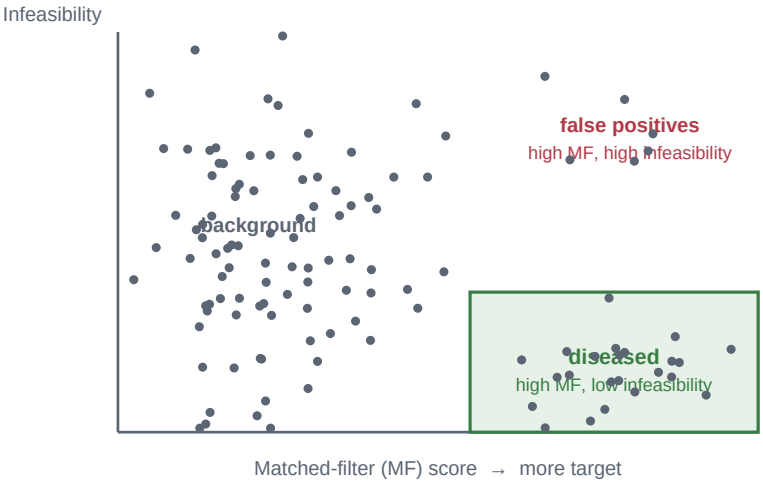


Figure 4. How each study obtained its endmembers. Endmember quality decides the result: the purest, most representative spectra must be used (Fitzgerald 2004).



Mixture tuned matched filtering (MTMF)

1. MNF transform separates signal from noise.
2. Only the TARGET spectrum (diseased crop) is needed; other endmembers stay unknown.
3. MF score = estimated target abundance *(Zhang 2014 equated it with disease index).*
4. Infeasibility = how unlikely the pixel is a real target-background mixture.
5. Keep pixels with high MF and low infeasibility (rule classifier thresholds).

Kumar 2012; Zhang 2014

Figure 5. MTMF: pixels with a high matched-filter score and low infeasibility are taken as diseased; infeasibility removes false positives. Schematic.

5. Methodology of each study

5.1 Fitzgerald et al. 2004: constrained LSU and MESMA for spider mite damage in cotton (AVIRIS, 18 m)

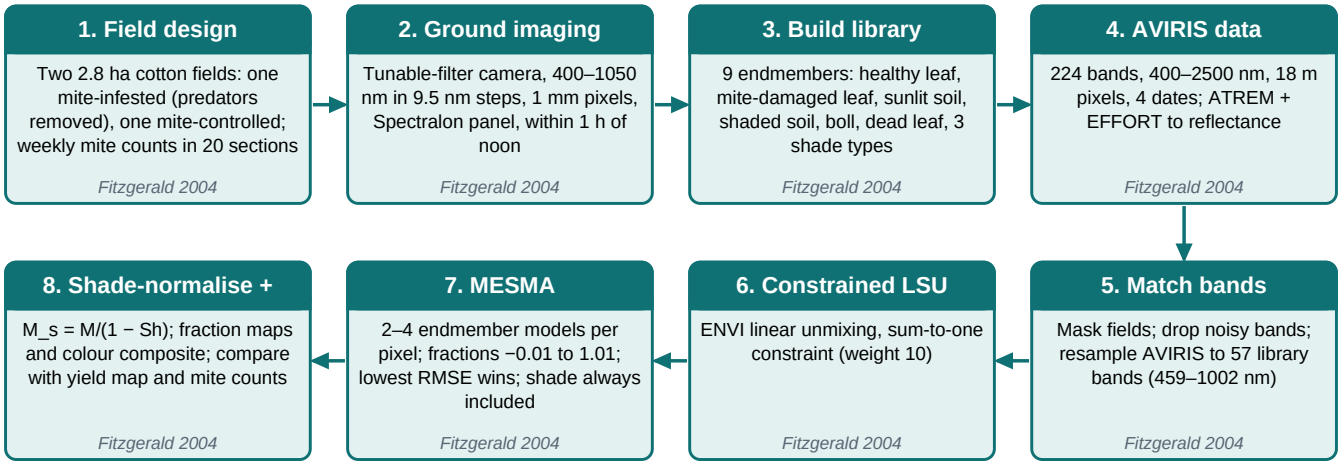
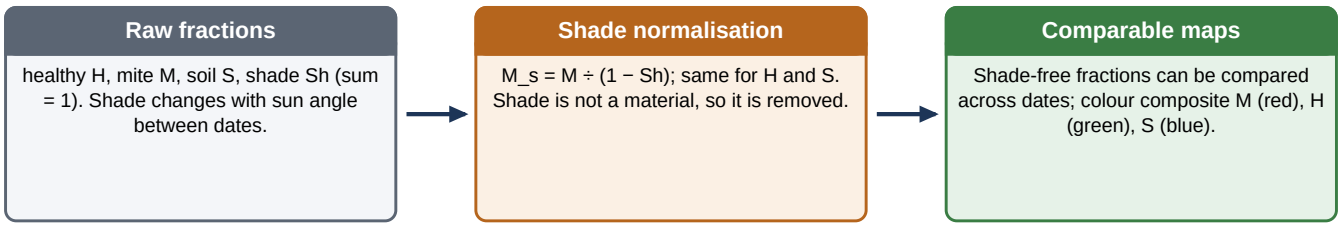


Figure 6. Fitzgerald et al. 2004 workflow.



Fitzgerald et al. 2004 (after Adams et al. 1993)

Figure 7. Shade normalisation makes fractions from different dates and sun angles comparable.

5.2 Kumar et al. 2012: MTMF, SAM and LSU for citrus greening (HLB) from aircraft

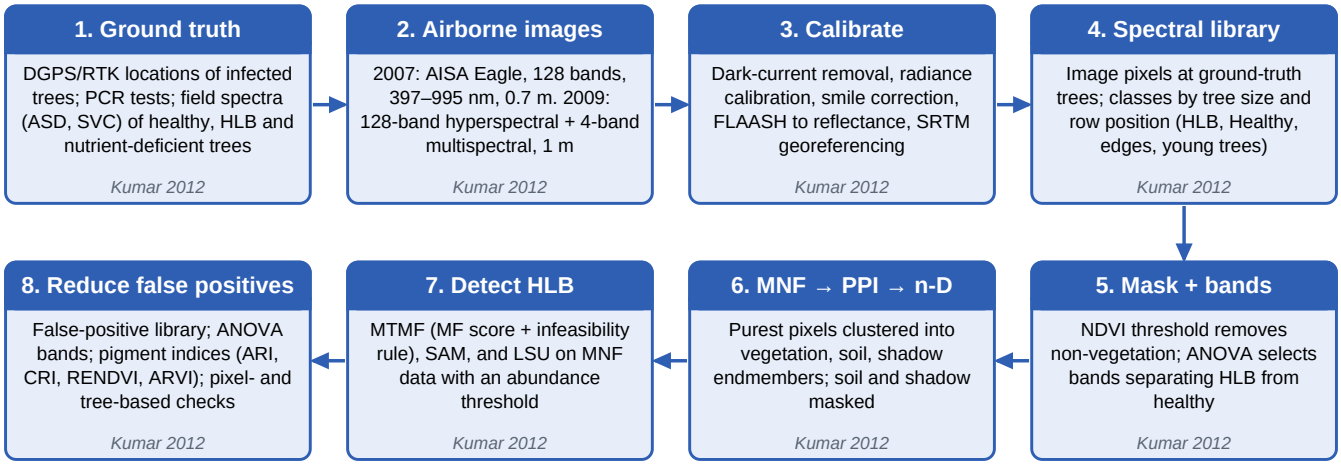


Figure 8. Kumar et al. 2012 workflow.

5.3 Zhang et al. 2014: MTMF with PLSR for wheat powdery mildew from 30 m satellite data

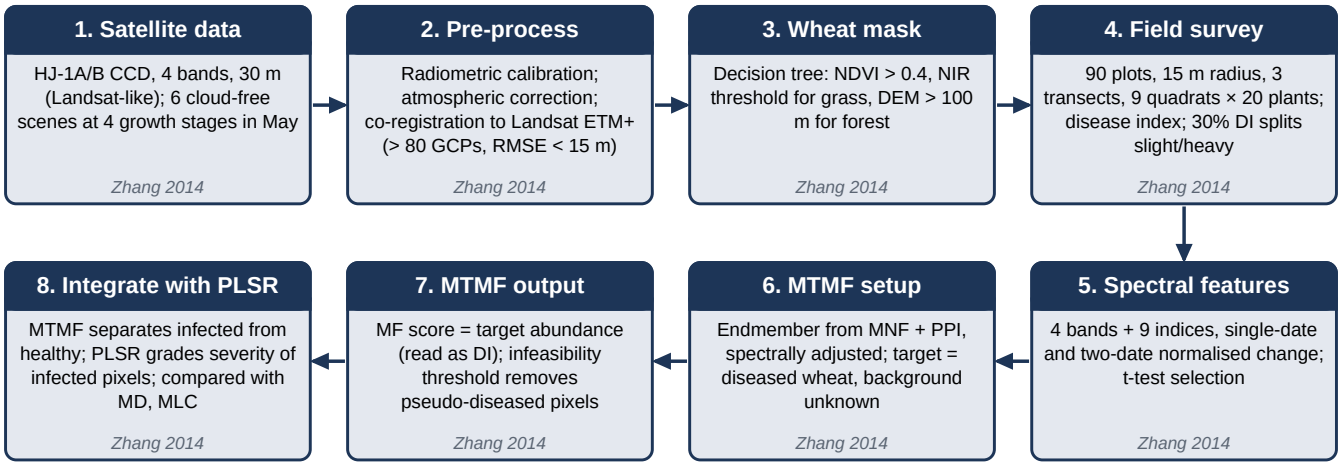


Figure 9. Zhang et al. 2014 workflow: unmixing applied to moderate-resolution satellite pixels.

5.4 Guan et al. 2025: superpixel unmixing for peanut leaf spot from UAV hyperspectral images

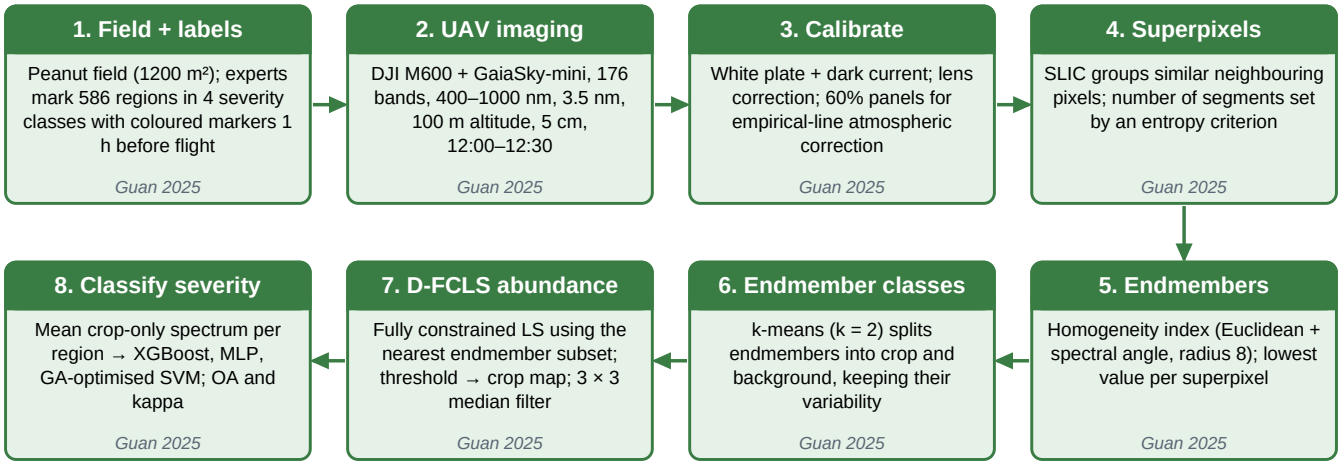


Figure 10. Guan et al. 2025 workflow: unmixing used to isolate crop spectra before disease classification.

5.5 Clenet et al. 2019: non-linear unsupervised unmixing for vine disease at three scales (conference abstract)

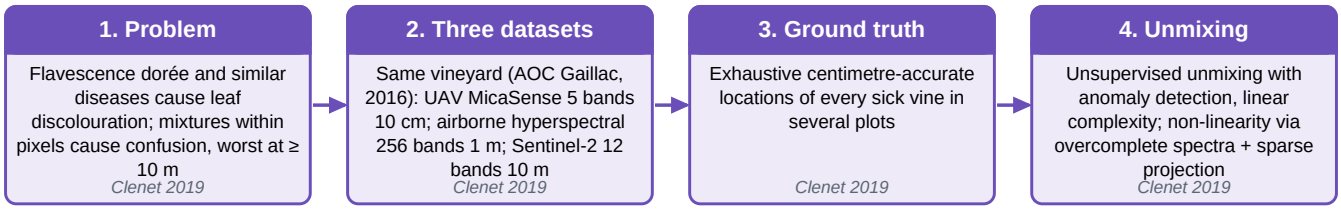


Figure 11. Clenet et al. 2019 method (Nakhostin et al. 2016 algorithm): endmembers for healthy and diseased leaves are found without a prior library.

6. Comparison of the unmixing designs

Study	Platform · pixel · bands	Endmember source	Unmixing model	Output used for disease
Fitzgerald 2004	AVIRIS · 18 m · 57 used	Ground tunable-filter library (9 spectra)	Constrained LSU + simplified MESMA	Shade-normalised mite fraction maps
Kumar 2012	Aircraft · 0.7–1 m · 128 (and 4)	Image library at ground-truth trees; MNF → PPI → n-D	MTMF, SAM, LSU	Thresholded MF score / abundance → infected pixels
Zhang 2014	HJ-1 CCD satellite · 30 m · 4	MNF + PPI, spectrally adjusted	MTMF (+ PLSR)	MF score as DI; infeasibility filter; severity by PLSR
Guan 2025	UAV · 5 cm · 176	Superpixel homogeneity index + k-means	D-FCLS (fully constrained)	Crop-only spectra → severity classifiers
Clenet 2019	UAV, aircraft, Sentinel-2 · 0.1–10 m · 5–256	Unsupervised (overcomplete dictionary)	Non-linear sparse unmixing + anomaly detection	Sub-pixel healthy vs sick leaf contributions

7. Applying unmixing to satellite mixed pixels

- **Few bands limit endmembers.** A 4-band sensor supports at most about 5 endmembers; MTMF suits such data because only the disease target must be specified (Zhang 2014).
- **Use pure endmembers from finer scales.** Ground or UAV spectra, resampled to the satellite's bands, give purer references than mixed satellite pixels (Fitzgerald 2004).
- **Mask non-crop first.** Crop masks remove spectral contrast between land covers that would swamp the disease signal (Zhang 2014; Kumar 2012).

- **Handle shade and sun angle.** Normalise for shade before comparing dates (Fitzgerald 2004).
- **Allow variability.** Use endmember classes or per-pixel subsets (MESMA, D-FCLS) rather than one spectrum per material (Fitzgerald 2004; Guan 2025).
- **Consider non-linear mixing.** At 10 m, leaf–soil multiple scattering matters; non-linear sparse methods are being tested for Sentinel-2 (Clenet 2019).
- **Check physical plausibility.** Strongly negative or > 1 fractions, high RMSE or high infeasibility mean an endmember is missing or wrong (Fitzgerald 2004; Kumar 2012).

8. References

Clenet, H., Nakhostin, S., Laroche-Pinel, E. & Duthoit, S. (2019). Non-linear unmixing as an innovative tool to detect vine diseases in UAVs, airborne and satellite images: preliminary results. *21st GiESCO International Meeting*, Thessaloniki, 23–28 June 2019, pp. 55–56.

Fitzgerald, G.J., Maas, S.J. & DeTar, W.R. (2004). Spider mite detection and canopy component mapping in cotton using hyperspectral imagery and spectral mixture analysis. *Precision Agriculture*, 5, 275–289. https://doi.org/10.1023/B:PRAG.0000032766.88787.5f

Guan, Q., Qiao, S., Feng, S. & Du, W. (2025). Investigation of peanut leaf spot detection using superpixel unmixing technology for hyperspectral UAV images. *Agriculture*, 15, 597. https://doi.org/10.3390/agriculture15060597

Kumar, A., Lee, W.S., Ehsani, R.J., Albrigo, L.G., Yang, C. & Mangan, R.L. (2012). Citrus greening disease detection using aerial hyperspectral and multispectral imaging techniques. *Journal of Applied Remote Sensing*, 6, 063542.

Zhang, J., Pu, R., Yuan, L., Wang, J., Huang, W. & Yang, G. (2014). Monitoring powdery mildew of winter wheat by using moderate resolution multi-temporal satellite imagery. *PLoS ONE*, 9(4), e93107. https://doi.org/10.1371/journal.pone.0093107

Method sources cited inside these papers: Adams et al. (1993) for shade normalisation; Roberts et al. (1998) for MESMA; Boardman et al. (1995) for MTMF; Nakhostin et al. (2016, IEEE TGRS) for the non-linear unmixing algorithm.